Abstract

Title: Quantum Machine Learning Classifier (QSVM)

Artificial Intelligence (AI) and Machine Learning (ML) have become an important part of many real-world applications, including healthcare, finance, cybersecurity, and image recognition. These technologies help in analysing large amounts of data and making accurate predictions. However, as datasets become larger and more complex, traditional machine learning algorithms often require more computational power and time, which can affect their performance. Quantum Computing is an emerging technology that offers a new approach to solving such computational challenges by using the principles of quantum mechanics. Combining quantum computing with machine learning has led to the development of **Quantum Machine Learning (QML)**, a field that has the potential to improve the efficiency of data analysis and classification.

The main objective of this project is to develop a **Quantum Machine Learning Classifier (QSVM)** and study how quantum computing can improve classification tasks when compared to traditional machine learning methods. In the existing system, **Support Vector Machines (SVMs)** use classical kernel functions to classify data. Although they perform well for many applications, they may struggle with complex, high-dimensional datasets and require higher computational resources. This creates the need for more advanced approaches that can represent data more effectively and improve classification performance.

To overcome these limitations, the proposed system uses a **Quantum Support Vector Machine (QSVM)**. Instead of relying only on classical kernel functions, the system uses **quantum feature mapping** to encode classical data into a quantum feature space. Quantum circuits are then used to calculate quantum kernels, which help identify similarities between data points more effectively. These quantum kernels are combined with a classical Support Vector Machine to build a hybrid quantum-classical model that utilizes the strengths of both technologies.

The project is implemented using **Python** with **Qiskit**, **PennyLane**, and **Scikit-learn**. **NumPy** and **Pandas** are used for data preprocessing, while **Matplotlib** is used to visualize the results. The quantum circuits are executed using **IBM Quantum Experience**, which provides access to quantum simulators and quantum hardware for testing the proposed model.

The overall workflow of the project begins with collecting a suitable dataset, followed by data preprocessing, including cleaning, normalization, and splitting the data into training and testing sets. The processed data is then encoded into quantum states using quantum feature maps. Quantum kernels are generated through quantum circuits and used to train the QSVM model. Finally, the performance of the model is evaluated using metrics such as **accuracy, precision, recall, F1-score**, and confusion matrix. The results are compared with those of a classical SVM to analyse the effectiveness of the proposed approach.

The expected outcome of this project is to demonstrate how Quantum Machine Learning can improve classification for complex datasets while providing a better understanding of quantum-based learning techniques. The project also serves as a practical learning platform that connects the concepts of quantum computing with modern machine learning. In the future, this approach can be extended to support more advanced applications in healthcare, finance, cybersecurity, and other data-intensive domains, contributing to the development of next-generation intelligent systems.